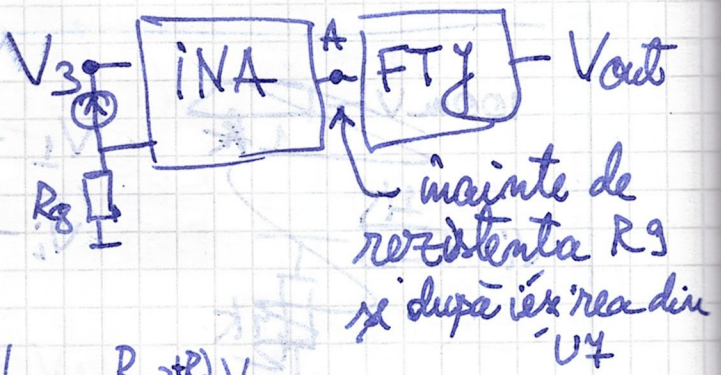
Problemă examen

Primul etaj → INA
Al doilea etal → Filtru trece jos

$$V_A = -\frac{R_6}{R_4}\left(1+\frac{R_2+R_3}{R_1}\right)\cdot V_3$$



$$V_{out} = V_A \cdot \left(1+\frac{R_{11}}{R_{12}}\right)$$

$$\Rightarrow V_{out} = -\left(1+\frac{R_{11}}{R_{12}}\right)\frac{R_6}{R_4}\left(1+\frac{R_2+R_3}{R_1}\right)V_3$$

a) $-10\cdot 10^{-3}\left(1+\frac{1K}{1K}\right)\frac{10K}{10K}\left(1+\frac{2K}{100}\right)$

$= -0{,}01 \cdot 2 \cdot 21 = -0{,}42 V_3$

b) $V_{out} = -2 \cdot 21\, V_3 = -42 V_3$

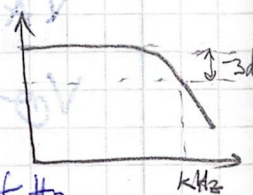
$V_{out} \in [V_{S-}, V_{S+}] \Rightarrow V_{3\,min} = \frac{12V}{-42} = -0{,}28$

$V_{3\,max} = \frac{-12V}{-42} = 0{,}28$

$\Rightarrow V_3 \in [-0{,}28,\ 0{,}28]$

c) domeniu de intrare de x2 mai mic ⇒ amplif x2 mai mare ⇒ $R_6 = 20K$

d) $f_c \sim \frac{1}{\sqrt{C_1 C_2 R_9 R_{10}}} \cdot ct$



$\sim \frac{1}{20\cdot 10^{-9}\cdot 2\cdot 10^{3}} = \frac{1}{4\cdot 10^{-5}} = 25\,KHz$

⇒ ordinul KHz

e)

